

David Guo

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EDUCATION

University of Toronto

M.A.Sc. in Mechanical and Industrial Engineering **Supervisor:** Prof. Scott Sanner

Sep 2026 - 2028 (expected)

University of Toronto

B.A.Sc. in Engineering Science (Machine Intelligence) **GPA:** 3.85/4.0, with Honours

2021–2026

Awards: U of T Scholar - \$7,500 (2021), Rowe Innovation Award - \$4,000 (2022), Dean's List (2022–2026)

RESEARCH EXPERIENCE

Data-Driven Decision Making (D3M) Lab, University of Toronto

May 2025 – Present

Researcher | Supervisor: Prof. Scott Sanner – Supported by NSERC USRA.

- Co-designed and conducted a human–human study in multimodal conversational recommendation, producing the VOGUE dataset with 60 dialogues, approximately 900 turns, and 1,440 item-level feedback annotations.
- Co-first-authored two peer-reviewed papers at ACM UMAP and Findings of ACL 2026; led the analysis, interpretation, manuscript development, and peer review revisions for VOGUE, and presented it at UMAP.
- Extended this work by developing LLM-based conversational and embodied recommender agents, combining user-state modeling, retrieval, planning, and multimodal interaction.

Matter Lab, University of Toronto

Sep 2025 – Sep 2026

Research Student | Supervisors: Dr. Ishrath Mohamed Irshadeen and Dr. Nasim Abdollahi

- Developed and implemented a composition-aware graph neural network for molecular solubility prediction across unary, binary, and arbitrary multisolvent systems, including generalization to previously unseen solutes.
- Built the end-to-end deep learning pipeline across multiple public datasets totaling approximately 200,000 measurements, spanning molecular graph construction, mixture representation, model training, and evaluation.
- Validated performance and generalization across more than 80 model configurations through ablation and out-of-distribution studies, while leading the analysis and manuscript development.

Acceleration Consortium, University of Toronto

Apr 2026 – Sep 2026

Machine Learning Research Student | Supervisor: Dr. Nasim Abdollahi

- Lead machine learning development for self-driving laboratory workflows, building graph neural network models and evaluation pipelines for nanoemulsion particle size prediction and active learning.

PUBLICATIONS

*Equal contribution

ACL '26 Findings Liang, J.^{*}, Liu, Y.^{*}, **Guo, D.^{*}**, Jiang, Y., Sun, M., & Sanner, S.
Evaluating Scene-based In-Situ Item Labeling for Immersive Conversational Recommendation.

UMAP '26 **Guo, D.^{*}**, Sun, M.^{*}, Jiang, Y.^{*}, Liang, J., & Sanner, S.
VOGUE: A Multimodal Conversational Fashion Recommendation Dataset.

INDUSTRY EXPERIENCE

ModiFace (L'Oréal) | Software Developer Intern – Augmented Reality

May 2024 – May 2025

- Helped re-architect and ship the Vue-based integration layer of ModiFace's AR makeup platform by tracing redundant camera and image augmentations across several legacy libraries and simplifying their integration.
- Contributed exception-handling, telemetry, and diagnostic improvements for production applications deployed across L'Oréal brands, supporting more reliable debugging and maintenance.

DealerVu | Software Developer

May 2023 – Aug 2023

- Shipped end-to-end internationalization for the SuperSOM dealership platform, implementing dynamic language loading, locale-aware formatting, adaptive React layouts, and supporting .NET/SQL infrastructure.

SELECTED PROJECTS

Normative Spinal Geometry Modeling for Metastatic Disease

Fall 2025

Capstone Research Project | Sunnybrook Research Institute

- Developed a hybrid framework combining gradient-boosted tree regression with statistical shape modeling to reconstruct patient-specific normative geometry across metastasis-affected regions, achieving under 6 mm 3D RMSE.
- Integrated the models into a Python library and visualization dashboard for assessing spinal metastasis-related deformity.

Pancreas Lesion Segmentation and Classification

Winter 2024/2025

- Extended nnU-Net V2 with a multi-task classification head for joint pancreas segmentation and lesion-subtype prediction.
- Achieved a pancreas Dice score of 0.92 and macro-average precision of 0.82 across lesion subtypes.

TECHNICAL SKILLS

Languages: Python, C/C++, C#, MATLAB, JavaScript/TypeScript, SQL
Scientific: NumPy, Pandas, SciPy, RDKit, OpenCV

ML: PyTorch, GNNs, LLMs, RAG, agentic AI, scikit-learn
Software: React, Vue, .NET, Git, Docker, Linux, LaTeX